

Lesson 5 Electric Potential

Date / /

Problem 35:

Given,

$$\vec{V} = 2 \left(\frac{V}{m^2} \right) x^2 - 3 \left(\frac{V}{m^2} \right) y^2$$

$$\vec{E} = ?$$

$$E_x = - \frac{\partial V}{\partial x}$$

$$= - \frac{\partial}{\partial x} \left[2 \left(\frac{V}{m^2} \right) x^2 - 3 \left(\frac{V}{m^2} \right) y^2 \right]$$

$$= - 2 \times 2x \left(\frac{V}{m^2} \right)$$

$$= - 4 \left(\frac{V}{m^2} \right) \cdot x$$

$$E_y = - \frac{\partial V}{\partial y}$$

$$= - \frac{\partial}{\partial y} \left[2 \left(\frac{V}{m^2} \right) x^2 - 3 \left(\frac{V}{m^2} \right) y^2 \right]$$

$$= - (-2) \times 3 \left(\frac{V}{m^2} \right) \cdot y$$

$$= + 6 \left(\frac{V}{m^2} \right) \cdot y$$

$\therefore$ we evaluate at $x = 3m$ and $y = 2m$ to obtain, $\vec{E} = E_x \hat{i} + E_y \hat{j}$



$$= -4 \left(\frac{V}{m^2} \right) \cdot x \hat{i} + \left\{ +6 \left(\frac{V}{m^2} \right) \cdot y \right\} \hat{j}$$

$$= \left\{ -4 \left(\frac{V}{m^2} \right) \cdot 3m \right\} \hat{i} + \left\{ +6 \left(\frac{V}{m^2} \right) \cdot 2m \right\} \hat{j}$$

$$\vec{F} = -12 \left(\frac{V}{m} \right) \hat{i} + 12 \left(\frac{V}{m} \right) \hat{j}$$

Problem 47:

The escape speed may be calculated from the requirement that the initial kinetic energy (of launch) be equal to the absolute value of the initial potential energy.

Thus, $K.E = P.E$

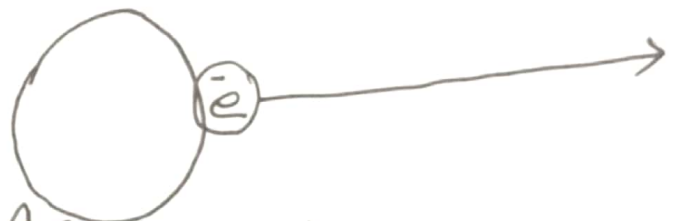
$$\Rightarrow \frac{1}{2} m v^2 = \frac{1}{4\pi\epsilon_0} \frac{q e}{r}$$

$$\Rightarrow v^2 = \frac{1}{4\pi\epsilon_0} \times \frac{2q e}{m r}$$

$$\Rightarrow v = \sqrt{\frac{9 \times 10^9 \times 2 \times 1.6 \times 10^{-15}}{(9.11 \times 10^{-31}) \cdot (0.01)}}$$

$$\therefore v = 22490 \text{ m/s}$$

$$\therefore v = 2.2 \times 10^4 \text{ m/s}$$



Given,

$$m = 9.11 \times 10^{-31} \text{ kg}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$q = 10000e$$

$$q = 1.6 \times 10^{-15} \text{ C}$$

$$r = 1 \text{ cm}$$

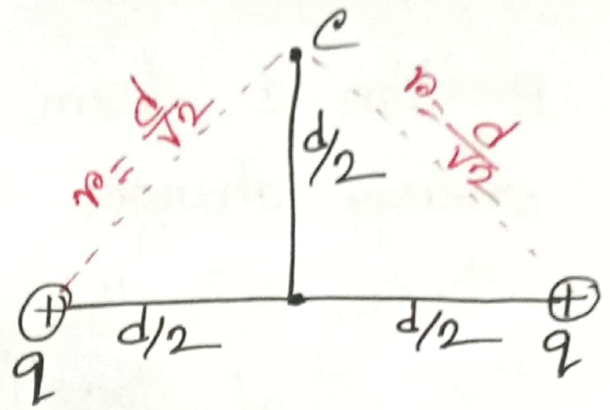
$$r = 0.01 \text{ m}$$

$$v = ?$$

Problem 88 :

Given, $q = 2 \mu\text{C}$
 $= 2 \times 10^{-6} \text{ C}$

$d = 2 \text{ cm}$
 $= 0.02 \text{ m}$



$$r = \sqrt{(d/2)^2 + (d/2)^2}$$

$$r = \frac{d}{\sqrt{2}}$$

(a) The electric potential at C is the sum of the potential due to the individual charges since they produce the same potential, it is twice that of either one.

$$V = 2 \times \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r}$$

$$\Rightarrow V = 2 \times \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{d/\sqrt{2}}$$

$$\Rightarrow V = 2 \times 9 \times 10^9 \cdot \frac{(2 \times 10^{-6}) \times \sqrt{2}}{0.02} \text{ V}$$

$$\Rightarrow V = 2.5 \times 10^6 \text{ V}$$



(b) As you move the charge into position c from far away the potential energy changes from zero to qV , where, V is the electric potential at the final location of the charge.

The change in the potential energy equals the work you must do to bring the charge in:

$$W = qV$$
$$\Rightarrow W = (2 \times 10^{-6} \text{ C}) \times (2.54 \times 10^6 \text{ V})$$

$$\therefore \boxed{W = 5.1 \text{ J}}$$

(c) The work calculated in part (b) represents the potential energy of the interactions between the charge brought in from infinity and the other two charges. To find the total potential energy of the three-charge system you must add the potential energy of the interaction between

the fixed charges. Their separation is d so that potential energy is $\frac{q^2}{4\pi\epsilon_0 d}$. Date / /

The total potential energy is,

$$U = W + \frac{q^2}{4\pi\epsilon_0 d}$$

$$\Rightarrow U = 5.1 \text{ J} + \frac{(9 \times 10^9)(2 \times 10^{-6})^2}{0.02 \text{ m}}$$

$$\Rightarrow U = (5.1 + 1.8) \text{ J}$$

$$\boxed{U = 6.9 \text{ J}}$$

